

41. The method of bisection is used to find the root of $\sin x + x - 1 = 0$ starting with the interval $[0, 2]$. After the first approximation, the interval which contains the root becomes $[0, 1]$. Find the interval which contains the root after the third approximation.
- A. $[0, 0.25]$
- B. $[0.25, 0.75]$
- C. $[0.5, 0.75]$
- D. $[0.5, 1]$
- E. $[0.75, 1]$
42. John goes to school and returns home at speeds x km/h and $(x + 1)$ km/h respectively. The school is 2 km from John's home and the total time for the two journeys is 54 minutes. Which of the following equations can be used to find x ?
- A. $\frac{x}{2} + \frac{x+1}{2} = \frac{54}{60}$
- B. $\frac{2}{x} + \frac{2}{x+1} = \frac{54}{60}$
- C. $\frac{\frac{1}{2}[x + (x+1)]}{4} = \frac{54}{60}$
- D. $\frac{4}{\frac{1}{2}[x + (x+1)]} = \frac{54}{60}$
- E. $2x + 2(x+1) = \frac{54}{60}$